

RAJALAKSHMI
ENGINEERING COLLEGE
An AUTONOMOUS Institution
Affiliated to ANNA UNIVERSITY, Chennai

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

(CYBER SECURITY)

CR23532- Cyber and Digital Forensics

LAB MANUAL

2023 REGULATION

FIFTH SEMESTER

III YEAR

EXPERIMENT 1

STUDY OF COMPUTER FORENSICS AND DIFFERENT TOOLS USED FOR FORENSIC INVESTIGATION

AIM:

The objective is to become familiar with computer forensics, learn the stages followed when investigating digital evidence, and review the tools commonly used for this work.

SOFTWARE REQUIREMENTS:

- Operating System: Windows / Linux
- Digital forensic tools: Autopsy, FTK Imager, The Sleuth Kit, and other forensic utilities.

THEORY:

Computer forensics, often called digital forensics, applies systematic methods to find, acquire, safeguard, inspect, and interpret information stored on digital devices. It supports inquiries into cybercrime, security breaches, unauthorized system use, and other incidents involving electronic systems.

Investigators may review hard drives and other storage media, operating-system records, application logs, web-browsing traces, deleted data, and related sources to locate information connected with a case.

A central requirement is to examine digital information without compromising its reliability, so that the evidence and the conclusions drawn from it can be documented and explained accurately.

DIGITAL FORENSICS PROCESS:

A typical digital investigation is carried out through the stages listed below:

1. Identification

Determine which devices or records may contain useful evidence, including computers, hard disks, USB media, mobile phones, and system logs.

2. Collection

Acquire the relevant data with suitable forensic methods and tools, following the required handling procedures.

3. Preservation

Keep acquired material safe from accidental damage or alteration. Creating a forensic image and recording hash values can help demonstrate that the evidence has remained unchanged.

4. Examination

Inspect the acquired material for useful files, metadata, deleted content, and other digital traces.

5. Analysis

Interpret the extracted artifacts, connect related events, look for unusual activity, and determine which findings are significant to the case.

6. Presentation

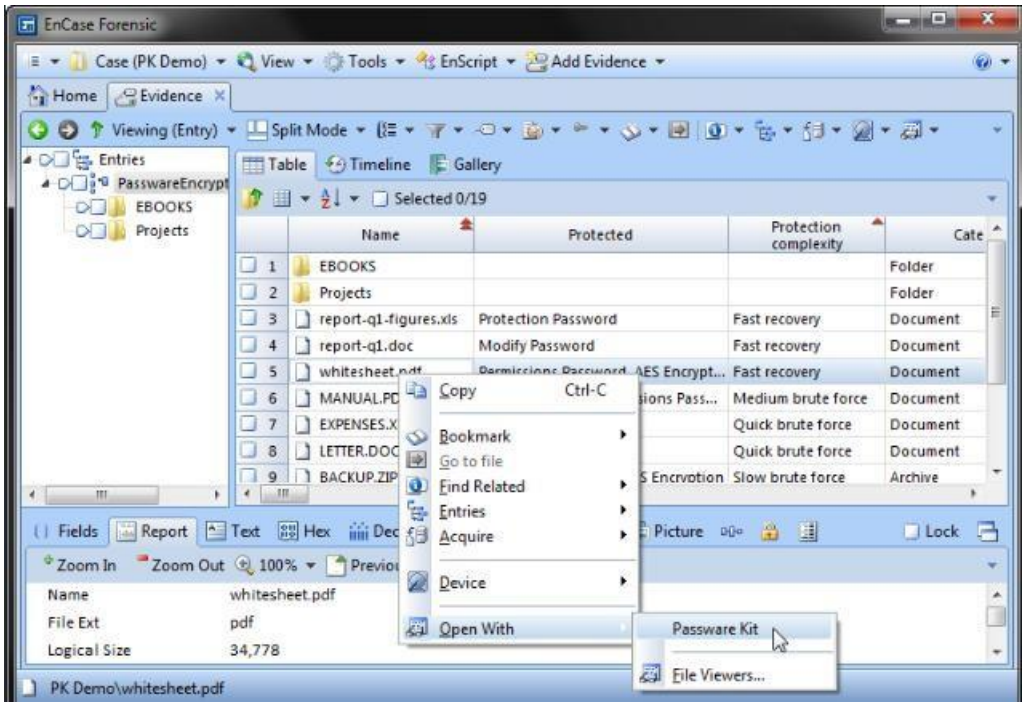
Prepare a clear report describing the procedures followed, the observations made, and the conclusions reached for technical or legal assessment.

DIFFERENT TOOLS USED IN COMPUTER FORENSICS

Digital investigations can be supported by both licensed forensic platforms and freely available or open-source utilities. The tools below are examples of these categories.

A. EnCase Forensic

1. Paid Tool: EnCase Forensic is a commercial digital investigation suite.
2. Evidence Acquisition: It helps examiners acquire digital evidence from computers and storage media.
3. Data Examination: It allows investigators to inspect files, folders, and other digital information.
4. Digital Investigation: It helps analyze evidence to identify relevant information related to an investigation.
5. Evidence Preservation: It supports maintaining the integrity of digital evidence during forensic examinations.



B. Forensic Toolkit (FTK)

1. Evidence Searching: FTK helps investigators search digital evidence quickly.
2. File Examination: It allows examiners to inspect files, folders, and their contents.
3. Data Analysis: It helps analyze information collected during a forensic examination.
4. Keyword Search: It supports searching for specific words or phrases in digital evidence.
5. Evidence Review: It helps investigators identify and review information relevant to the investigation.

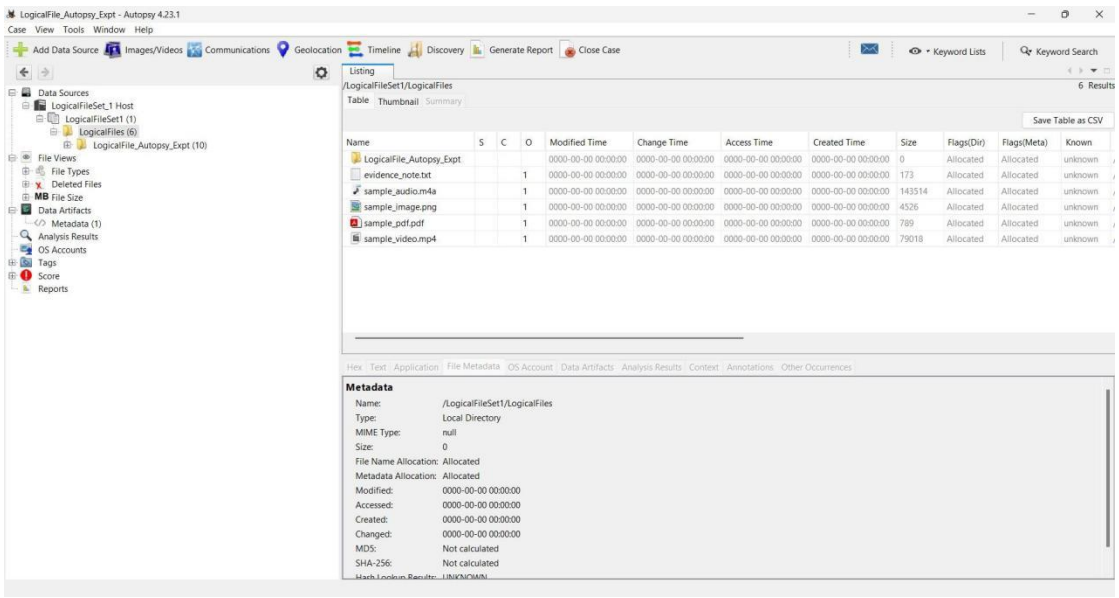
C. The Sleuth Kit (TSK)

1. Open-Source Tool: TSK is a free digital forensics toolkit.
2. Command-Line Tools: It provides utilities for forensic investigations.
3. Disk Analysis: It examines disk images and file systems.
4. Evidence Recovery: It helps locate files and data useful as evidence.

D. Autopsy

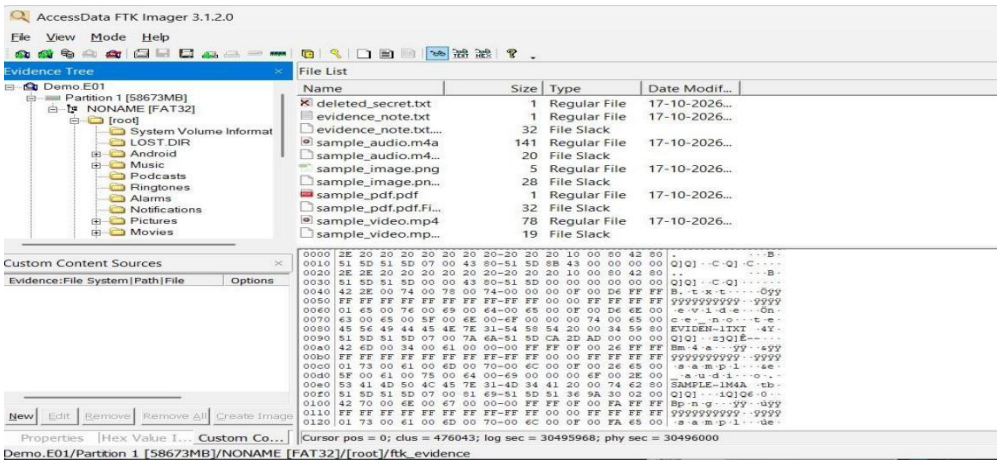
1. Graphical Interface: Provides a user-friendly workspace for forensic investigations.
2. Disk Analysis: Examines disk images and digital evidence.
3. Keyword Search: Helps find specific words or phrases.
4. File Recovery: Recovers deleted files where possible.
5. Artifact Analysis: Helps examine digital artifacts for relevant evidence.

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E. FTK Imager

1. Evidence Acquisition: Creates forensic copies of storage media.
2. Image Inspection: Examines forensic images of digital evidence.
3. File Viewing: Allows users to view files and folders within evidence images.
4. Evidence Preservation: Helps preserve the original evidence for investigation.



APPLICATIONS OF COMPUTER FORENSICS:

Common uses of digital forensics include the following:

1. Cybercrime inquiries: Reviewing electronic traces linked to intrusions, malicious software, and related online offences.
2. Data restoration: Attempting to retrieve files that have been deleted or can no longer be accessed.
3. Security incident analysis: Studying an event to understand how it may have occurred and what systems were affected.
4. Workplace inquiries: Reviewing relevant digital records as part of an appropriately authorized investigation into employee conduct.
5. Legal cases: Maintaining and examining electronic evidence that may be required in court or other formal proceedings.
6. Network analysis: Using connection logs and related records to investigate unusual communication or network behaviour.

GENERAL PROCEDURE:

1. Study the fundamental concepts of computer and digital forensics.
2. Identify the major stages of the digital forensic investigation process.
3. Study the different commercial and open-source forensic tools.
4. Understand the purpose and functions of tools such as FTK Imager, Autopsy, and The Sleuth Kit.
5. Examine the applications of digital forensics in cybercrime investigation and data recovery.
6. Document the observations and findings.

OBSERVATION:

The exercise introduced the role of computer forensics and showed why digital records can be important when examining cybercrime and other technology-related incidents.

The utilities were reviewed in terms of their main functions. The study also covered the sequence of a forensic examination: locating evidence, acquiring it, protecting it, inspecting and interpreting it, and reporting the findings.

RESULT:

Hence, the basic principles of computer forensics and the functions of several investigation tools were studied successfully.

EXPERIMENT 2

LIVE FORENSICS CASE INVESTIGATION USING AUTOPSY

AIM:

To carry out a basic examination in Autopsy by setting up a case, importing an evidence source, processing its contents, and reviewing the artifacts identified by the application.

SOFTWARE REQUIREMENTS:

- Operating System: Windows
- Software: Autopsy Digital Forensic Tool
- Evidence source: A suitable forensic image or disk image
- Hardware: Computer with sufficient storage and memory

THEORY:

1. Definition: Autopsy is a graphical application used for examining digital evidence.

2. Integration: It incorporates The Sleuth Kit (TSK) and other forensic utilities.

3. File-System Analysis: It helps examine files, folders, and disk images.

4. Keyword Search: It allows investigators to search for specific words or phrases.

5. Deleted File Recovery: It attempts to recover deleted files where possible.

6. Case Creation: An examiner creates a case to organize case details and digital evidence.

7. Evidence Acquisition: The examiner adds an evidence source, such as a forensic disk image, to the case.

8. Ingest Modules: Relevant modules are selected to process and analyze the evidence.

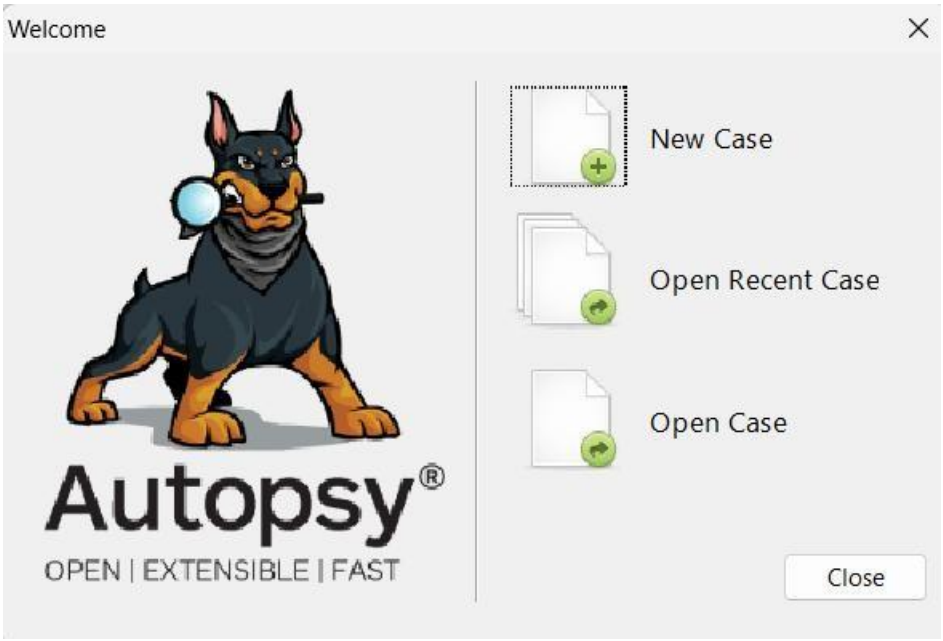
9. Artifact Analysis: It presents files, browser history, account-related information, metadata, and other digital artifacts for examination.

10. Investigation: The results help investigators identify information relevant to the digital forensic investigation.

PROCEDURE:

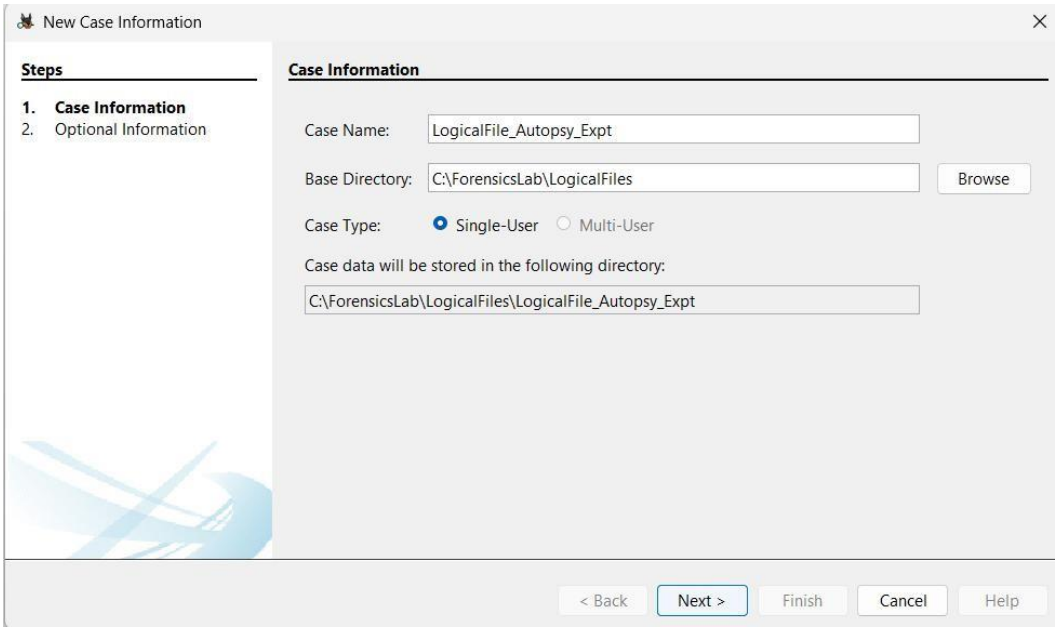
Open Autopsy

1. Obtain Autopsy from its official website and complete the installation.
2. Start Autopsy on the computer.
3. On the opening screen, choose New Case.



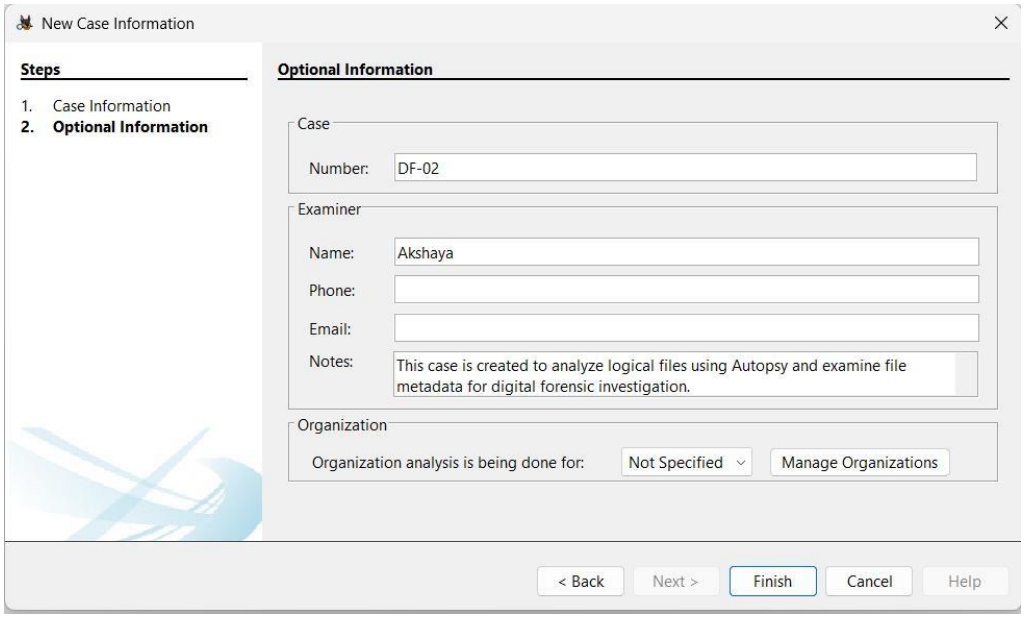
Create a new case

1. Type an appropriate name for the case.
2. Choose the base folder in which Autopsy should save the case files.
3. Select Next to continue.



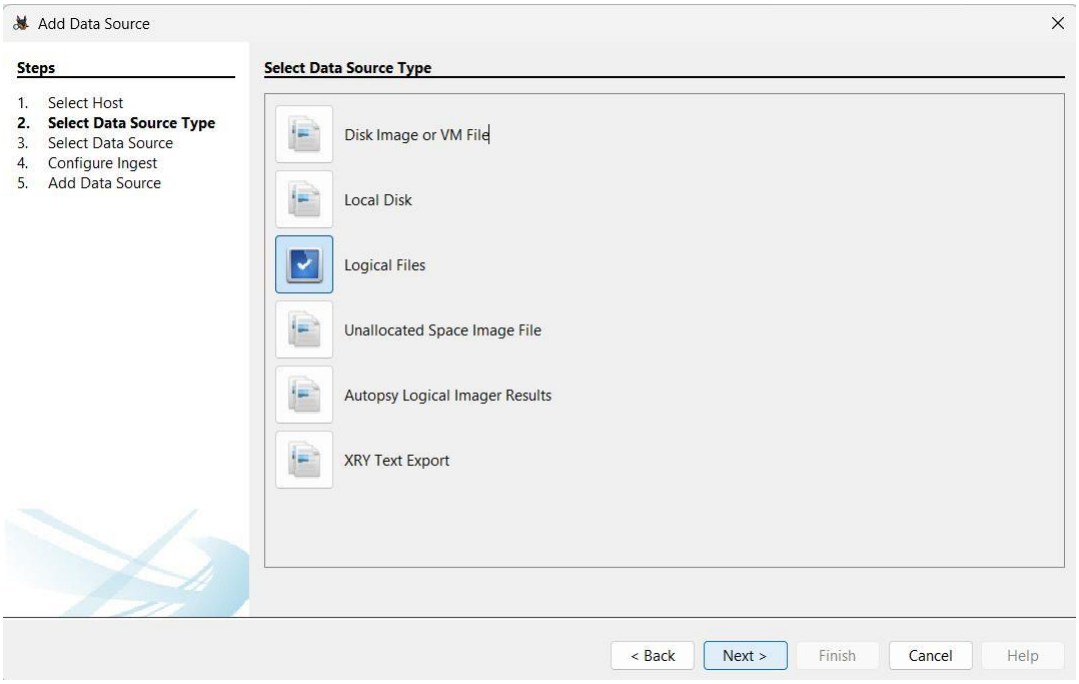
Enter case information

1. Fill in the case identification number.
2. Enter the details of the person conducting the examination.
3. Complete any other case fields requested by the program.
4. Choose Finish to save and create the case.



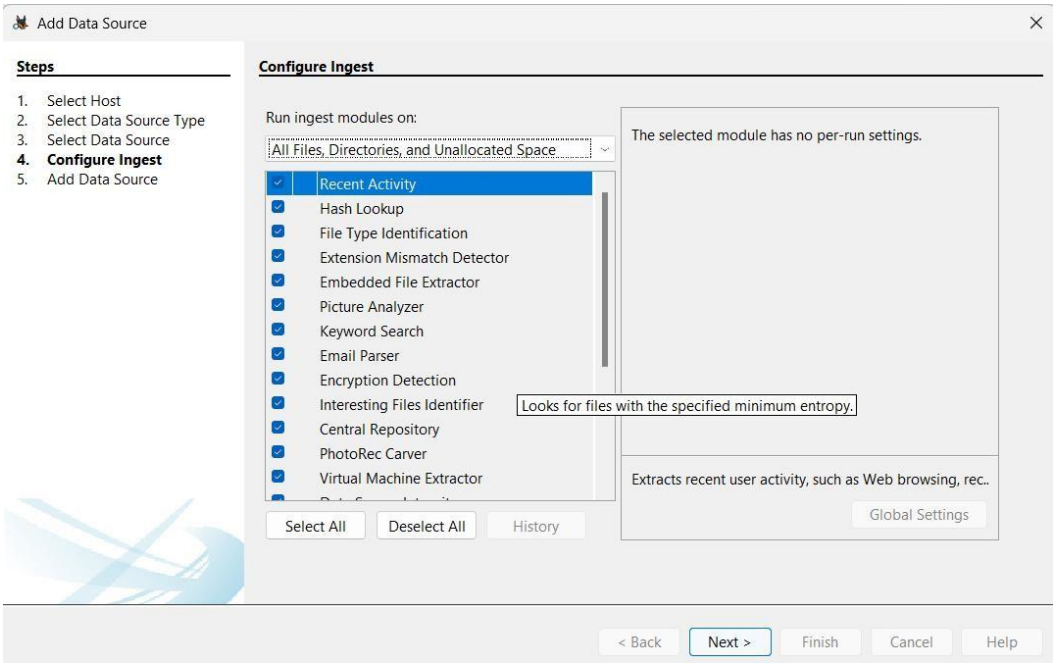
Add a data source

1. Once the case has been created, choose the option for adding evidence.
2. Select the relevant source category, for example a disk image or virtual-machine file.
3. Navigate to the folder containing the forensic image.
4. Choose the required image and continue.



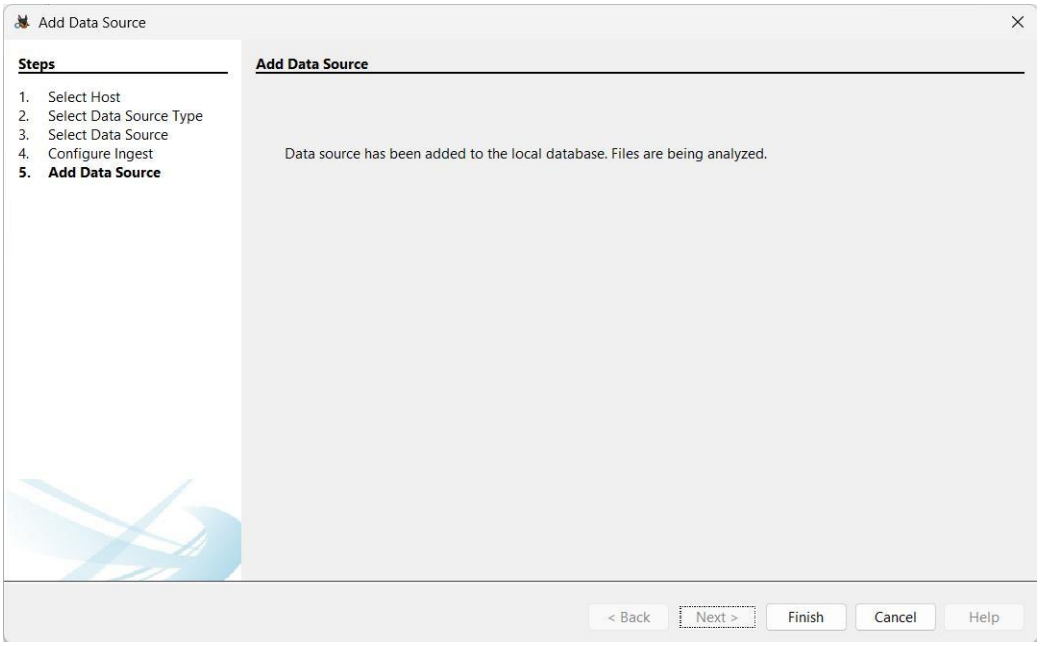
Configure ingest modules

1. Look through the ingest modules offered by Autopsy.
2. Enable the modules appropriate to the examination.
3. Press Next to move to the following screen.
4. Review the settings and complete the data-source addition.



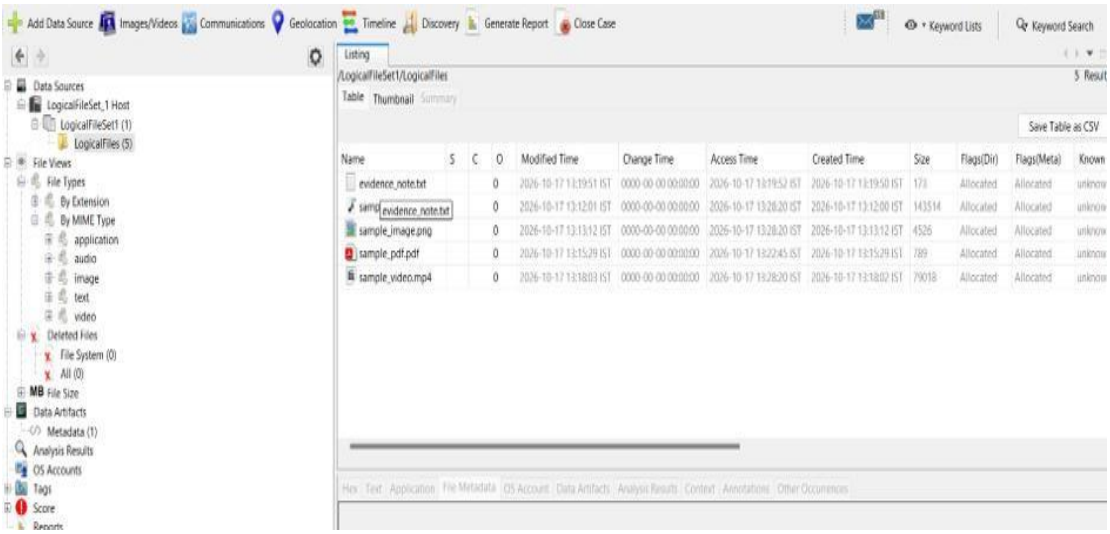
Process the data source

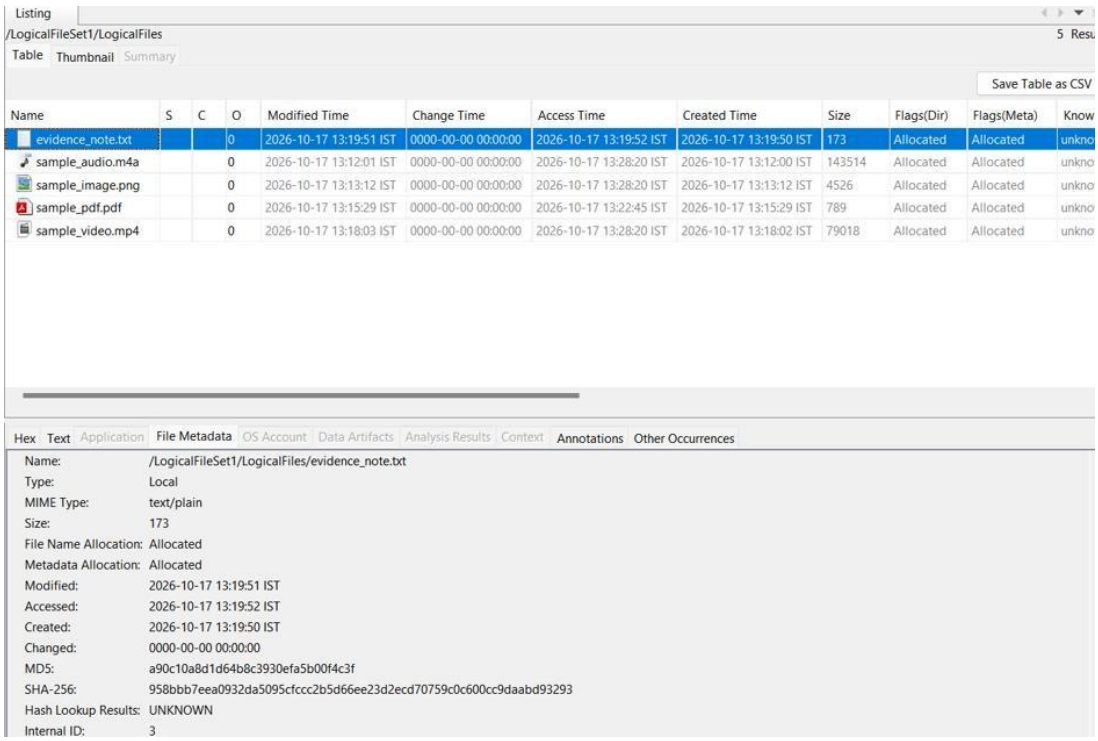
1. Finish the Add Data Source wizard.
2. Let Autopsy analyse the chosen image.
3. Allow the processing task to run to completion.
4. Review the information presented in the case window.



Examine the forensic artifacts

After processing, examine the available artifacts in Autopsy. Depending on the evidence and the modules selected.





VERIFICATION OF EVIDENCE INTEGRITY USING SHA-256:

Once the logical files had been processed in Autopsy, their integrity was checked again by calculating SHA-256 hash values through Windows Command Prompt.



OBSERVATION:

A new case was prepared in Autopsy and an evidence source was imported. The chosen ingest modules ran against the source and produced information that could be examined.

The case window displayed the files and artifacts detected during processing. The items available for review varied according to the data contained in the selected image.

RESULT:

Hence, an Autopsy-based forensic workflow was carried out by creating a case, importing and processing a data source, and reviewing the artifacts returned by the application.

EXPERIMENT 3

RECOVER DELETED FILES USING FORENSIC TOOLS

AIM:

To investigate a forensic image for deleted files, attempt recovery with suitable tools, and check the contents of any file retrieved.

SOFTWARE REQUIREMENTS:

- Operating System: Windows
- FTK Imager
- Recover My Files
- USB storage device or suitable test evidence source

THEORY:

1. File Deletion: Deleting a file does not necessarily erase its data permanently from the storage medium.

2. Space Allocation: The file system may mark the deleted file's storage space as available for future use.

3. Data Recovery: If new data has not overwritten the deleted file's storage space, some or all of its original content may be recovered.

4. Forensic Tools: Digital forensic utilities examine storage devices and forensic images to identify traces of deleted files.

5. FTK Imager: It is used to create forensic images of storage media and view their contents during an investigation.

6. Recovery Software: Specialized recovery tools search available data for deleted files that may be restored.

7. Evidence Preservation: A forensic image records the source data as it existed at the time of acquisition.

8. Safe Examination: Investigators analyze the forensic copy instead of repeatedly handling the original device, helping preserve the integrity of the original evidence.

PROCEDURE:

Create a sample file

1. Start Notepad.

2. Enter a short piece of test text.
3. Save the file as deleted_secret.txt on a test USB drive or another location.

```
CONFIDENTIAL – FORENSIC EXPERIMENT FILE

Case Reference: CF-2026-001
Project: Cyber and Digital Forensics
Prepared By: Student Name
Date: 17-10-2026

Subject: Sample confidential project document

This file contains simulated evidence for the Autopsy and FTK Imager
laboratory experiments.

Employee ID: EMP-10293
Username: student_forensics
Email: student.forensics@example.com
Project Code: CYBER-DFIR-2026
Access Level: Restricted

Confidential project notes:
The internal audit identified unauthorized access attempts on the
company research server. The suspect may have transferred project
documents to a USB storage device.

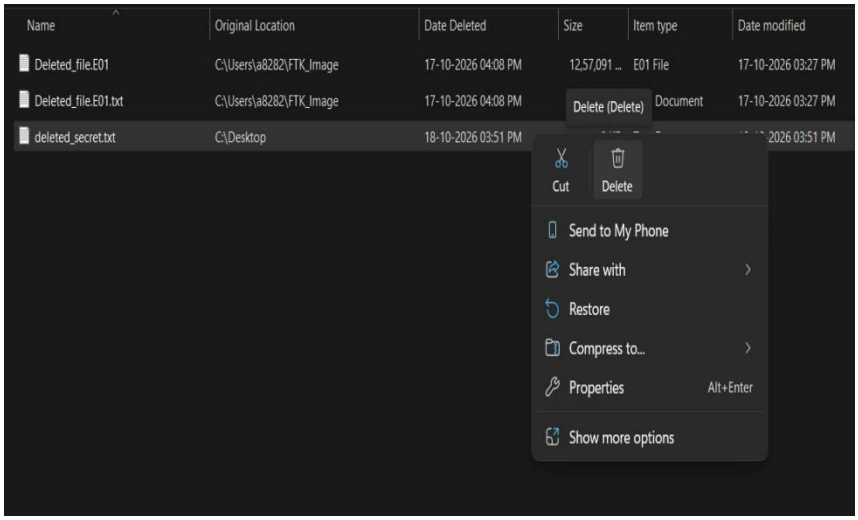
Important keywords:
confidential
unauthorized access
USB transfer
project documents

Password Hint: ForensicsLab@2026

Disclaimer:
This file contains fictional information created only for educational
and forensic analysis purposes.
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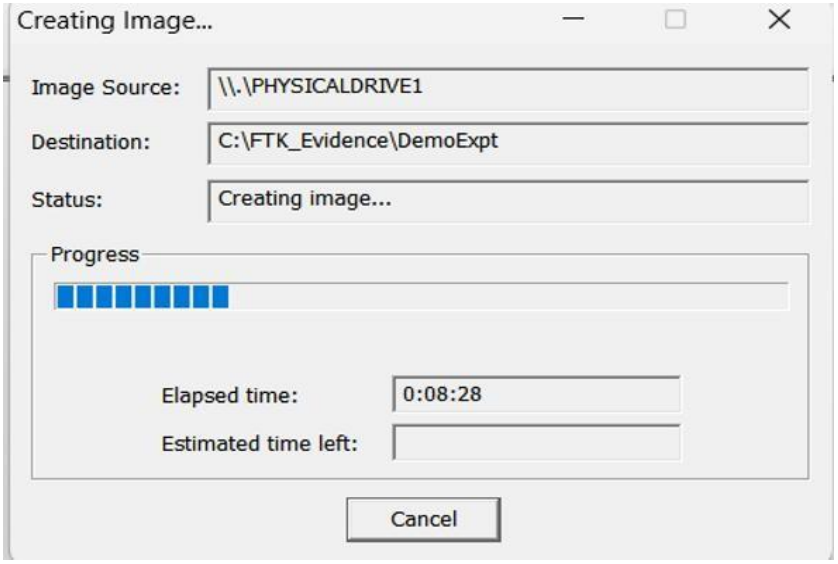
Delete the sample file

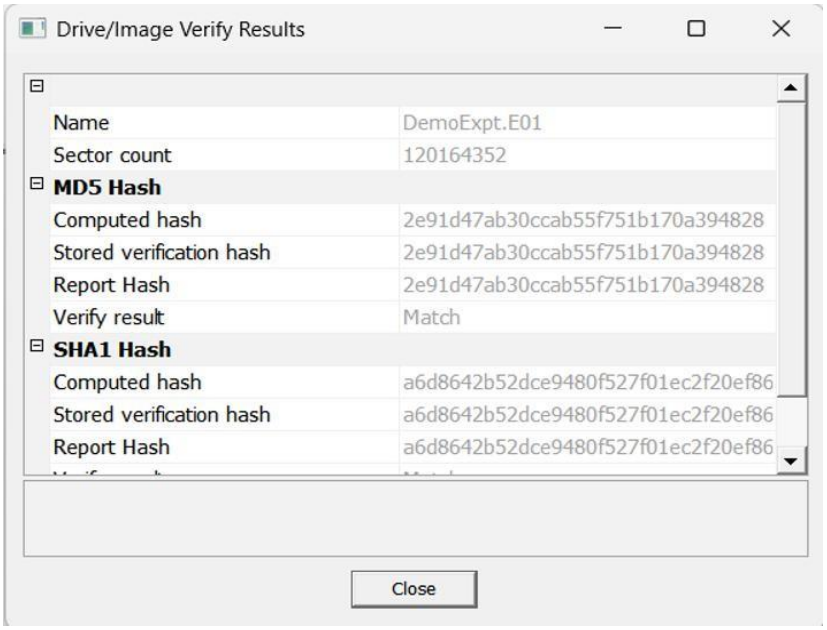
1. Find the file named deleted_secret.txt.
2. Use the right-click menu for the file and choose Delete.
3. Confirm that the file is no longer visible in its original location.

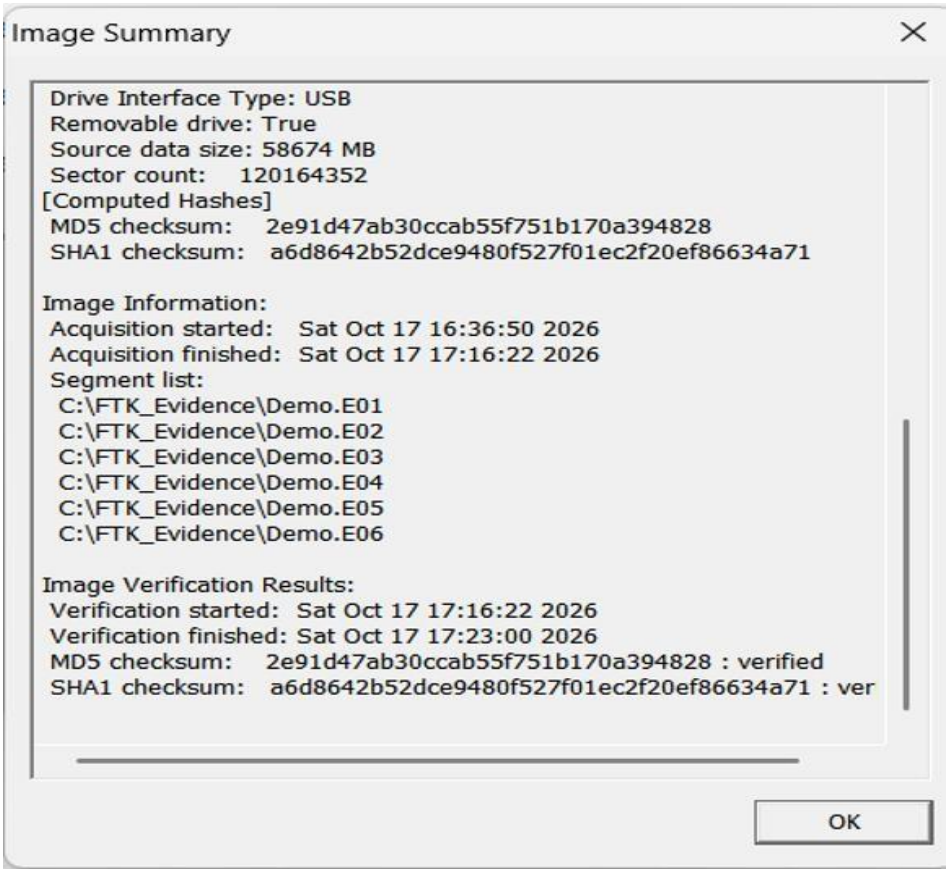


Create a forensic image using FTK Imager

1. Start the FTK Imager application.
2. From the File menu, choose Create Disk Image.
3. Select the source type that matches the test device.
4. Choose the test storage device on which the file was created.
5. Fill in the requested case and evidence information.
6. Set the folder for saving the image and select its format.
7. Begin acquisition and wait until the imaging operation finishes.

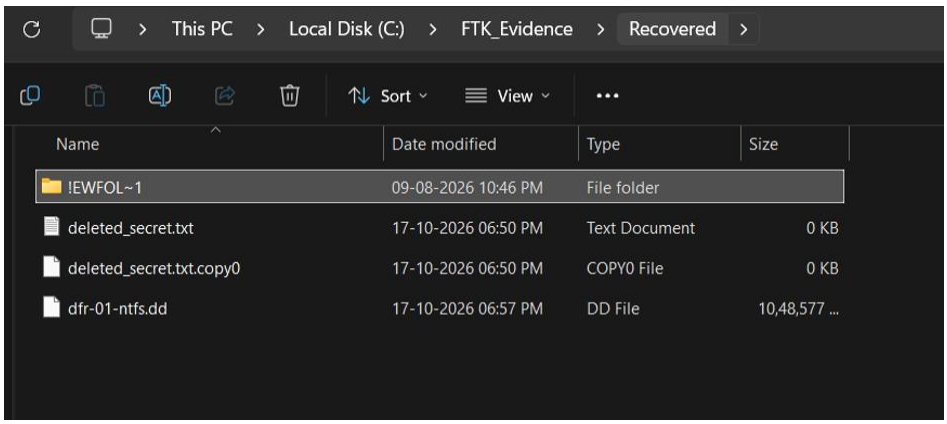






Verify the recovered file

1. Open the file produced by the recovery process.
2. Confirm whether the text entered in the sample file can be seen.
3. Compare the retrieved text with the original sample content.



OBSERVATION:

A test text document was saved to a storage location and deleted. FTK Imager was then used to make an image of the test source, which was checked with recovery software.

The output was checked to see if the deleted document and its text could be restored. Success depended on factors such as the file system, the acquisition process, and whether other data had replaced the deleted content.

RESULT:

Hence, a deleted-file recovery workflow was practised using forensic utilities. The image was searched for remnants of deleted data, and any recovered file was checked against the original sample where recovery was possible.

EXPERIMENT 4

FIND LAST CONNECTED USB ON YOUR SYSTEM (USB FORENSICS)

AIM:

To review the computer's USB connection history and inspect the identifying details recorded for devices by using USBDeview.

SOFTWARE REQUIREMENTS:

- Operating System: Windows
- Tool: USBDeview
- Computer with a history of USB device connections

THEORY:

1. Definition: USB forensics examines traces left when USB devices are connected to a computer.

2. USBDeview: It is a Windows utility used to view currently connected and previously used USB devices.

3. Device Name: Displays the name and description of the USB device.

4. Device Type: Identifies the type of USB device.

5. Serial Number: Shows the device's serial number, when available.

6. Connection Details: Displays the date and time the device was connected or added to the system, when recorded.

7. Vendor ID (VID): Identifies the device manufacturer.

8. Product ID (PID): Identifies the specific product model associated with the vendor.

9. Investigation: These details help investigators examine USB usage history and identify removable devices used on a computer.

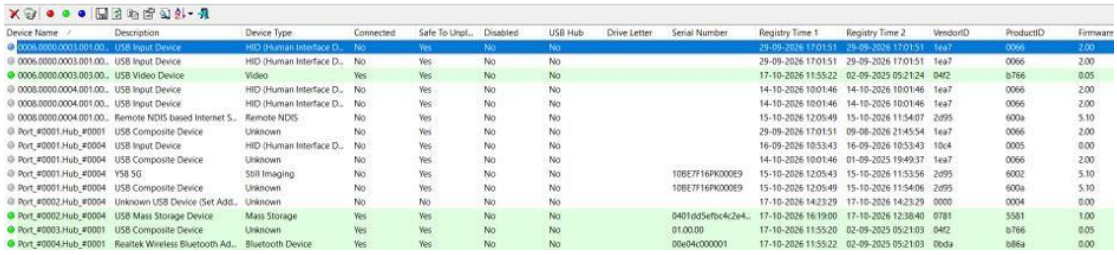
PROCEDURE:

1. Obtain USBDeview and run it on the Windows computer.

2. Wait for the program to populate its list of detected USB devices.
3. Review both the active-device entries and records of devices used earlier.
4. Locate the relevant device by its displayed name and description.
5. Check the serial number, VID, and PID fields when those values are available.
6. Review any connection or installation date-and-time entries shown for the device.
7. Write down the device details relevant to the examination.
8. Take screenshots of the full device list and the selected entry's properties.

OBSERVATION:

The utility presented current and historical USB entries, together with the identifying fields available for each device. These records can be used to select connections that may be relevant to the investigation.



Device Name	Description	Device Type	Connected	Safe To Unpl...	Disabled	USB Hub	Drive Letter	Serial Number	Registry Time 1	Registry Time 2	VendorID	ProductID	Firmware
0006.0000.0003.001.00...	USB Input Device	HID (Human Interface D...	No	Yes	No	No			29-09-2026 17:01:51	29-09-2026 17:01:51	1ea7	0066	2.00
0006.0000.0003.001.00...	USB Input Device	HID (Human Interface D...	No	Yes	No	No			29-09-2026 17:01:51	29-09-2026 17:01:51	1ea7	0066	2.00
0006.0000.0003.003.00...	USB Video Device	Video	Yes	Yes	No	No			17-10-2026 11:55:22	02-09-2025 05:21:24	04f2	b766	0.05
0008.0000.0004.001.00...	USB Input Device	HID (Human Interface D...	No	Yes	No	No			14-10-2026 10:01:46	14-10-2026 10:01:46	1ea7	0066	2.00
0008.0000.0004.001.00...	USB Input Device	HID (Human Interface D...	No	Yes	No	No			14-10-2026 10:01:46	14-10-2026 10:01:46	1ea7	0066	2.00
0008.0000.0004.001.00...	Remote NDIS based Internet S...	Remote NDIS	No	Yes	No	No			15-10-2026 12:05:49	15-10-2026 11:54:07	2d95	600a	5.10
Port_#0001.Hub_#0001	USB Composite Device	Unknown	No	Yes	No	No			29-09-2026 17:01:51	09-08-2026 21:45:54	1ea7	0066	2.00
Port_#0001.Hub_#0004	USB Input Device	HID (Human Interface D...	No	Yes	No	No			16-09-2026 10:53:43	16-09-2026 10:53:43	10c4	0005	0.00
Port_#0001.Hub_#0004	USB Composite Device	Unknown	No	Yes	No	No			14-10-2026 10:01:46	01-09-2025 19:49:37	1ea7	0066	2.00
Port_#0001.Hub_#0004	Y58 5G	Still Imaging	No	Yes	No	No		10BE7F16PK000E9	15-10-2026 12:05:43	15-10-2026 11:53:56	2d95	6002	5.10
Port_#0001.Hub_#0004	USB Composite Device	Unknown	No	Yes	No	No		10BE7F16PK000E9	15-10-2026 12:05:49	15-10-2026 11:54:06	2d95	600a	5.10
Port_#0002.Hub_#0004	Unknown USB Device (Set Add...	Unknown	No	No	No	No			17-10-2026 14:23:29	17-10-2026 14:23:29	0000	0004	0.00
Port_#0002.Hub_#0004	USB Mass Storage Device	Mass Storage	Yes	Yes	No	No		0401dd5efbc4c2e4...	17-10-2026 16:19:00	17-10-2026 12:38:40	0781	5581	1.00
Port_#0003.Hub_#0001	USB Composite Device	Unknown	Yes	Yes	No	No		01.00.00	17-10-2026 11:55:20	02-09-2025 05:21:03	04f2	b766	0.05
Port_#0004.Hub_#0001	Realtek Wireless Bluetooth Ad...	Bluetooth Device	Yes	Yes	No	No		00e04c000001	17-10-2026 11:55:22	02-09-2025 05:21:03	0bda	b86a	0.00



Properties

Device Name:	0006.0000.0003.003.000.000.000.000.000	Description:	USB Video Device
Device Type:	Video	Connected:	Yes
Safe To Unplug:	Yes	Disabled:	No
USB Hub:	No	Drive Letter:	
Serial Number:		Registry Time 1:	17-10-2026 11:55:22
Registry Time 2:	02-09-2025 05:21:24	VendorID:	04f2
ProductID:	b766	Firmware Revision:	0.05
WCID:		USB Class:	0e
USB SubClass:	03	USB Protocol:	00
Hub / Port:		Computer Name:	ADMIN
Vendor Name:		Product Name:	
ParentId Prefix:		Service Name:	usbvideo
Service Description:	@usbvideo.inf,%USBVideo.SvcDesc%;USB Vide	Driver Filename:	usbvideo.sys
Device Class:		Device Mfg:	Microsoft
Friendly Name:	HP Wide Vision HD Camera	Power:	500 mA
USB Version:	2.01	Driver Description:	USB Video Device
Driver Version:	10.0.26100.9711	Driver InfSection:	USBVideo.NT
Driver InfPath:	usbvideo.inf	Instance ID:	USB\VID_04F2&PID_B766&MI_00\7&28651dbc&(
Capabilities:	Removable, SilentInstall, SurpriseRemovalOK	Install Time:	
First Install Time:		Connect Time:	
Disconnect Time:			

OK

RESULT:

Hence, USBDeview was used to review the system's USB history and record the device information available for forensic analysis.

EXPERIMENT 5

VIEW LAST ACTIVITY OF YOUR PC

AIM:

To inspect activity traces and system-event information available on a Windows computer with the LastActivityView utility.

SOFTWARE REQUIREMENTS:

- Operating System: Windows
- Tool: LastActivityView

THEORY:

Definition: LastActivityView is a Windows utility used to view recent computer activity.

Activity Records: It collects activity records from different Windows data sources.

Unified Log: It displays available activity information in a single log.

System Events: It shows certain system events and recent user activities.

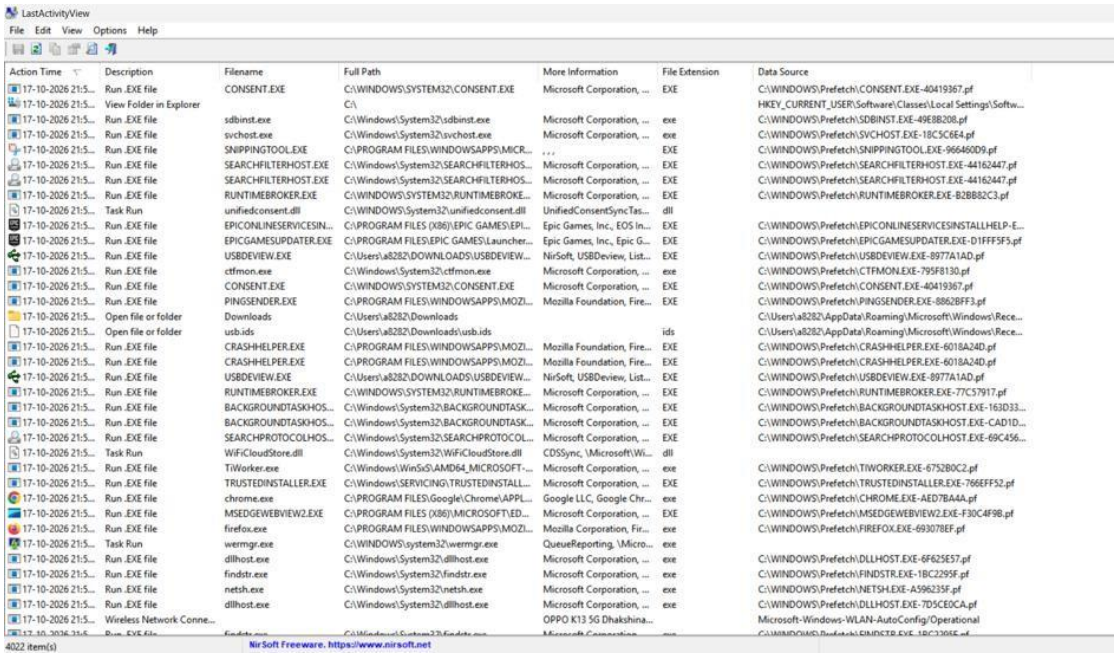
Investigation: It helps investigators review available computer activity records during digital forensic analysis.

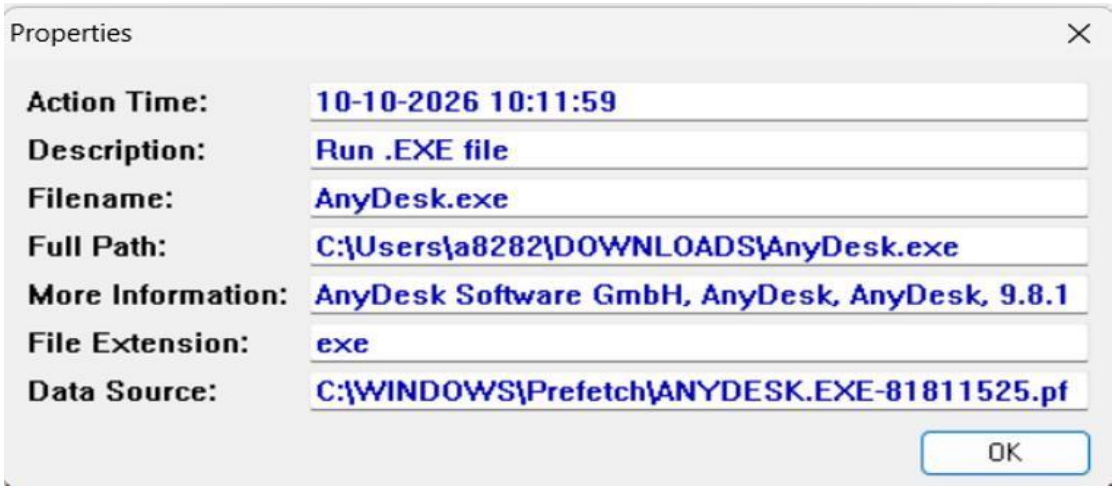
PROCEDURE:

1. Download and start LastActivityView on the Windows machine.
2. Let the utility collect the activity records it can access.
3. Inspect the log shown in the main application window.
4. Observe the available details, such as the activity description, filename or application, and recorded date and time, where available.
5. Identify relevant entries for recent computer activities.
6. Document the relevant findings for later analysis.
7. Save screenshots of the log and any entries of interest.

OBSERVATION:

The application displayed the activity entries it could retrieve from the Windows system. Reviewing these entries provided a view of some recent events, subject to the limits of the available records.





RESULT:

Hence, LastActivityView was used to inspect the system's accessible activity and event records for forensic purposes.

EXPERIMENT 6

EXTRACTION OF BROWSER ARTIFACTS

AIM

To review two categories of Google Chrome artifacts—cached resources and visited-page history—using ChromeCacheView and ChromeHistoryView.

SOFTWARE REQUIREMENTS

- Operating System: Windows
- Web Browser: Google Chrome
- Tools: ChromeCacheView and ChromeHistoryView

THEORY

1. Definition: Browsers store traces such as cached files and browsing history that may help in digital investigations.

2. ChromeCacheView: Reads and displays cached items stored by Google Chrome.

3. Source URL: Shows the URL associated with a cached item, when available.

4. Cache Details: Displays item type, cache filename, size, and access time.

5. ChromeHistoryView: Reads and displays recorded Chrome browsing-history entries.

6. Browsing Details: Shows website URLs, page titles, visit counts, and the most recent visit time, when available.

7. Investigation: Both tools help examiners analyze cached content and browsing history to understand a user's browsing activity.

4. PROCEDURE

A. EXAMINATION USING CHROMECACHEVIEW

1. Start ChromeCacheView on the Windows system after obtaining the utility.
2. Allow the program to find and load the Chrome cache records.

3. Review the cached-item list shown by the application.
4. Observe the available details, including the URL, file type, cache filename, file size, and access time.
5. Mark any cache entries that may be relevant.
6. Take a screenshot of the displayed cache records.

B. EXAMINATION USING CHROMEHISTORYVIEW

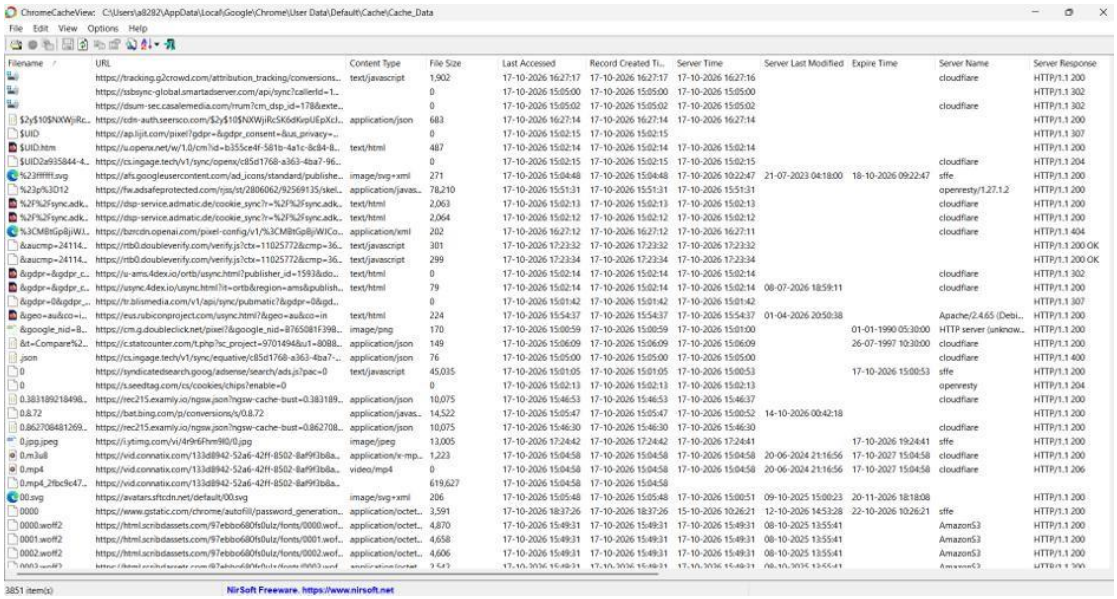
1. Open ChromeHistoryView on the Windows computer.
2. Let the utility load the available Chrome history database.
3. Review the history entries presented in the main window.
4. Observe the available details, including the URL, page title, visit count, and last visit time.
5. Select the browsing records of interest.
6. Capture the history list as a screenshot for the record.

OBSERVATION:

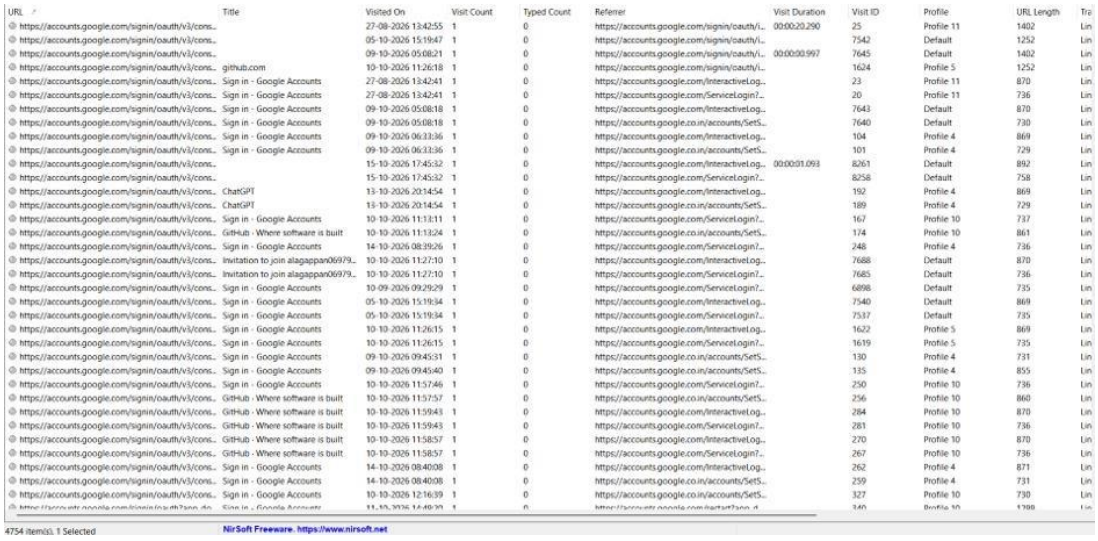
Each utility was used for a separate browser-data category: cached items and browsing-history entries.

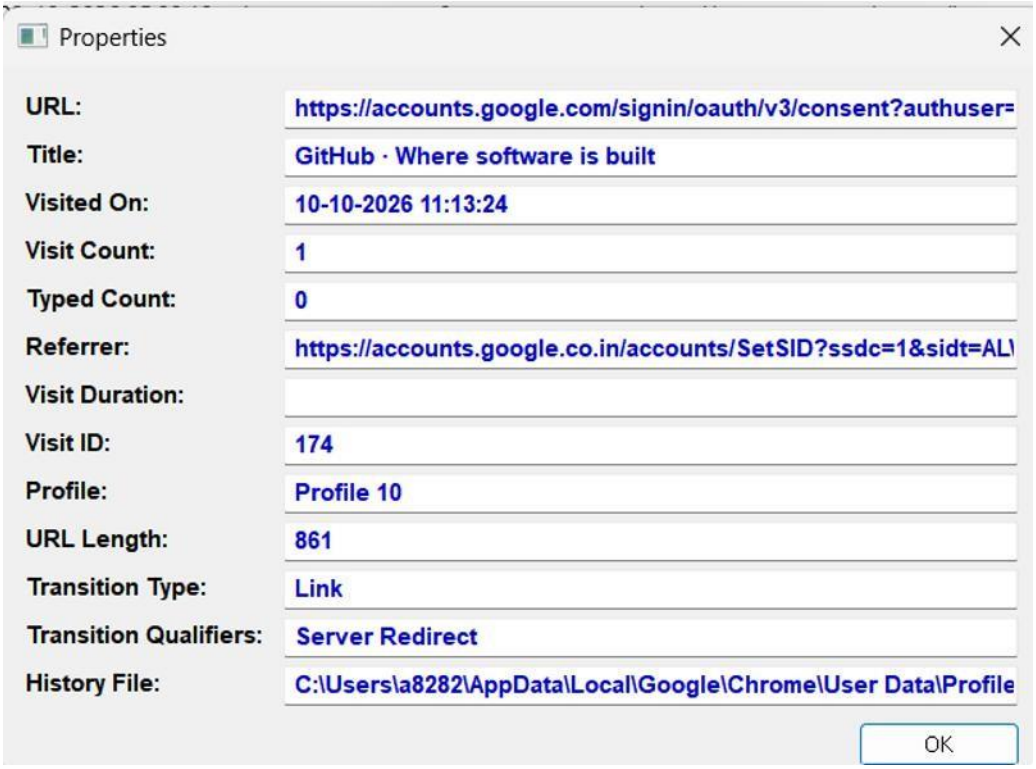
S. NO.	TOOL NAME	ARTIFACT EXAMINED	OBSERVATION
1	ChromeCacheView	Cached files	Cached file details were displayed where available.
2	ChromeHistoryView	Browsing history	Website history records were displayed where available.

DETAILS OBSERVED USING CHROMECACHEVIEW



DETAILS OBSERVED USING CHROMEHISTORYVIEW





RESULT:

Hence, ChromeCacheView and ChromeHistoryView were used to inspect browser-related evidence, including cached-resource details and the browsing-history records available on the system.

EXPERIMENT 7

COMPARISON OF TWO FILES FOR FORENSIC INVESTIGATION USING COMPARE IT SOFTWARE

AIM:

To use Compare It to examine a pair of files side by side and determine what content differs between them.

SOFTWARE REQUIREMENTS:

- Operating System: Windows
- Software: Compare It
- Two sample files for comparison

THEORY:

1. Definition: Compare It is a tool used to compare two versions of a file.
2. Side-by-Side Comparison: Displays both files in parallel panes.
3. Difference Detection: Highlights additions, deletions, and modifications.
4. File Analysis: Helps identify changes between document versions.
5. Forensic Investigation: Helps examiners detect and document possible file modifications.

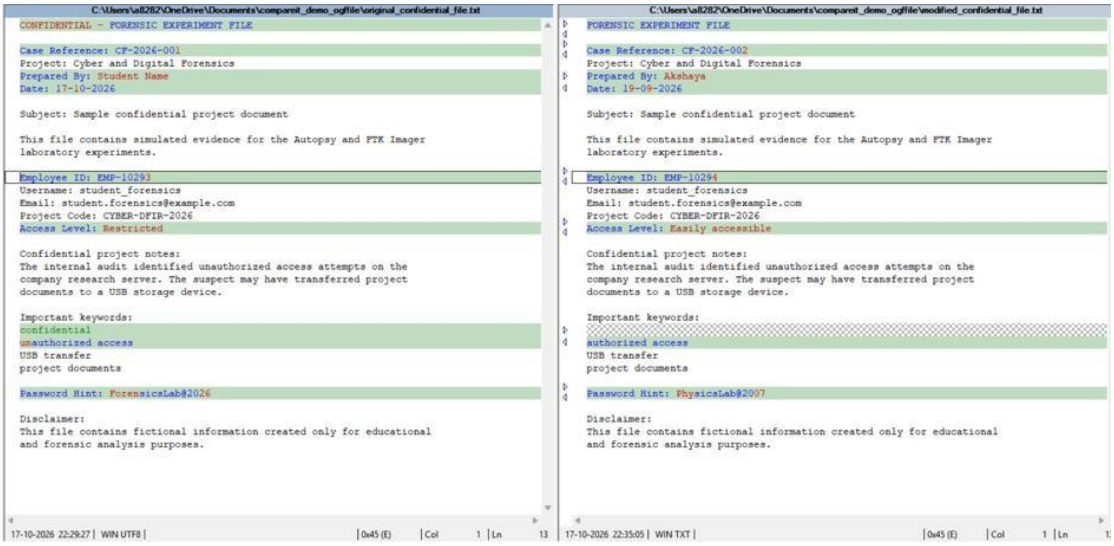
PROCEDURE:

1. Install Compare It on the Windows system and open it.
2. Create or select two test files that are mostly alike but include some differences.
3. Start the Compare It program.
4. Browse to the first file and choose OPEN.
5. Load the second file by selecting it and clicking OPEN.

6. Wait for the program to place the two files in adjacent panes.
7. Inspect the marked passages to see where the files do not match.
8. Note text that has been inserted, removed, or changed.
9. Write down the differences and take a screenshot of the comparison window.

OBSERVATION:

The two selected files appeared next to each other, with non-matching sections highlighted for easier review.





RESULT:

Hence, Compare It was used to compare two files and locate differences that may be relevant to a forensic examination.

EXPERIMENT 8

COLLECTING EMAIL EVIDENCE IN VICTIM PC

AIM:

To acquire a Windows computer's RAM with DumpIt and examine the memory output for artifacts that may relate to email activity.

SOFTWARE REQUIREMENTS:

- Operating System: Windows
- Tool: DumpIt
- Memory analysis tool or scripts specified in the lab procedure

THEORY:

1. Definition: Email forensics examines email-related traces to investigate communication activity.
2. Memory Acquisition: DumpIt captures a physical memory dump from a Windows system.
3. Volatile Data: RAM may contain temporary data from running applications, including traces of browser-based email activity.
4. Memory Analysis: Forensic tools analyze the memory dump to locate relevant email artifacts.
5. Artifact Identification: Investigators may find email addresses and other information related to email usage.
6. Evidence Preservation: The memory image and analysis results must be stored securely and documented to maintain evidence integrity.

PROCEDURE:

A. CAPTURING RAM USING DUMPIT

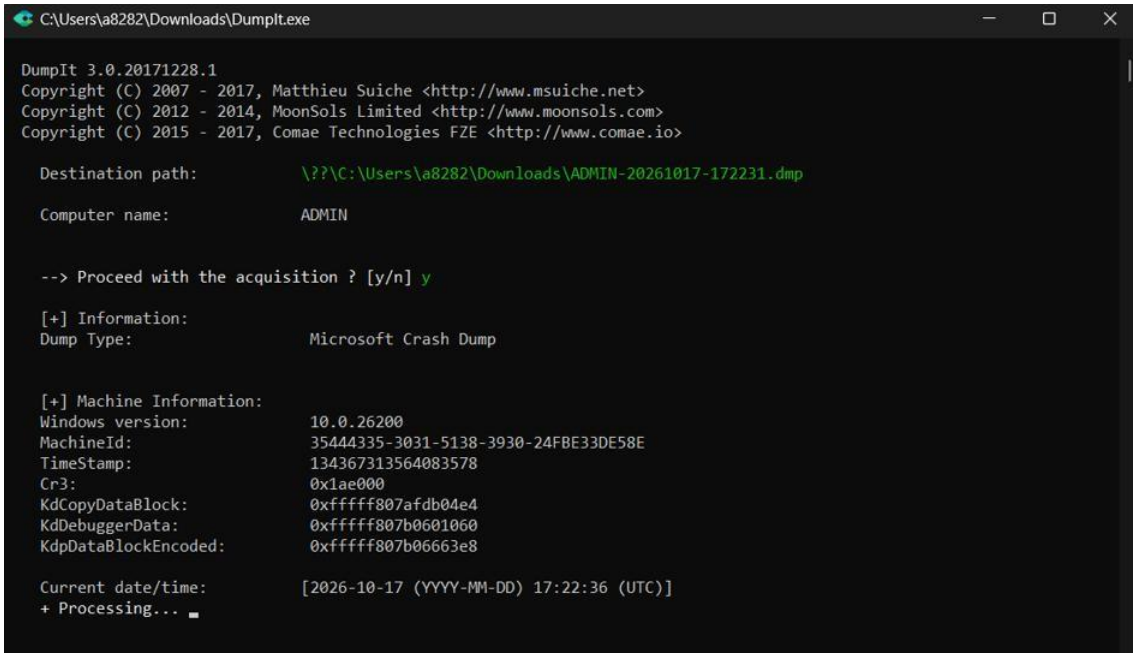
1. Place the DumpIt utility on the Windows computer and prepare it for use.
2. Execute DumpIt.exe with administrator privileges as required.

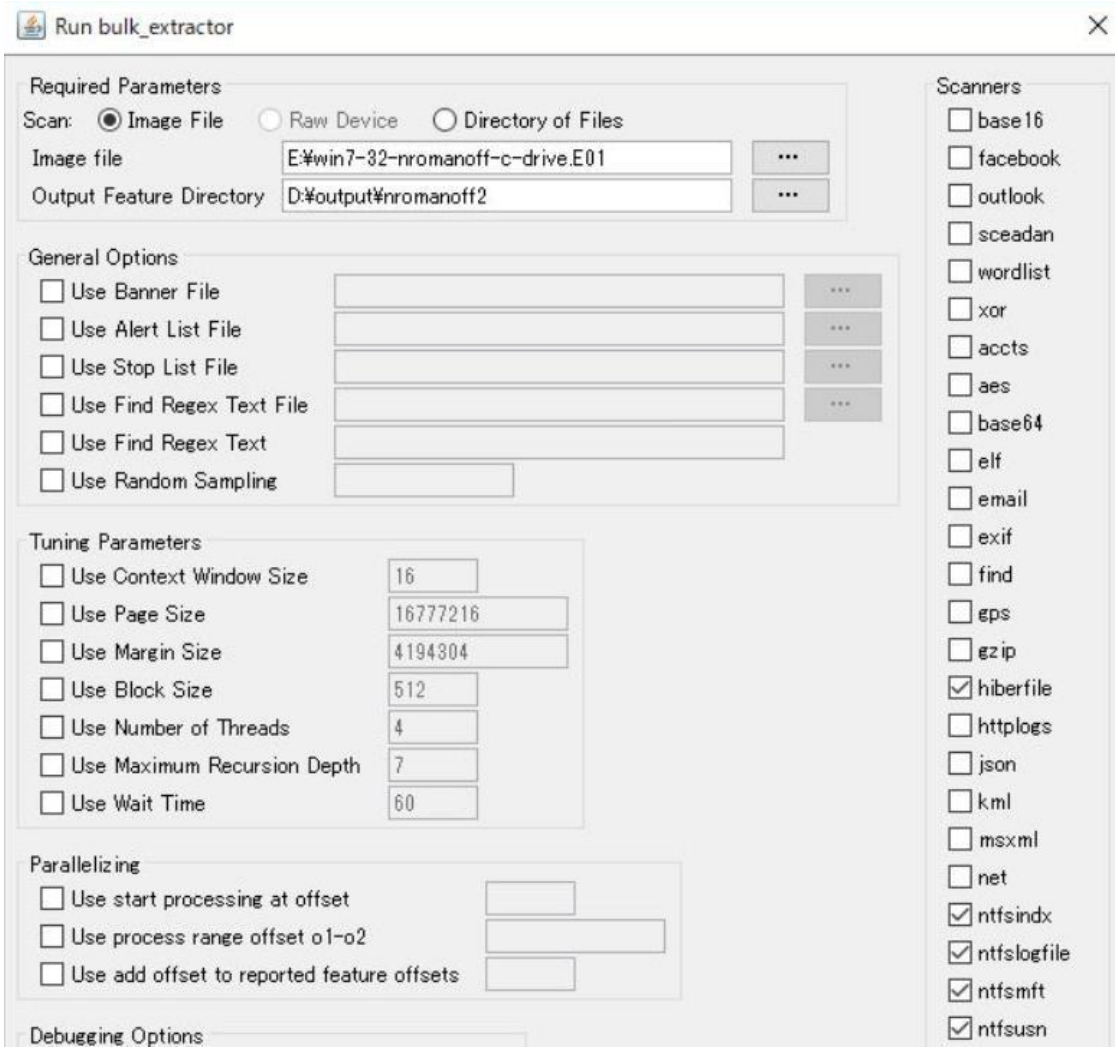
3. Review the confirmation message shown by DumpIt.
4. As directed by the manual, type Y to approve and start capturing memory.
5. Do not proceed until the memory capture has finished.
6. Find the raw dump file in the output location indicated by the utility.
7. Note the dump's name and size for documentation.

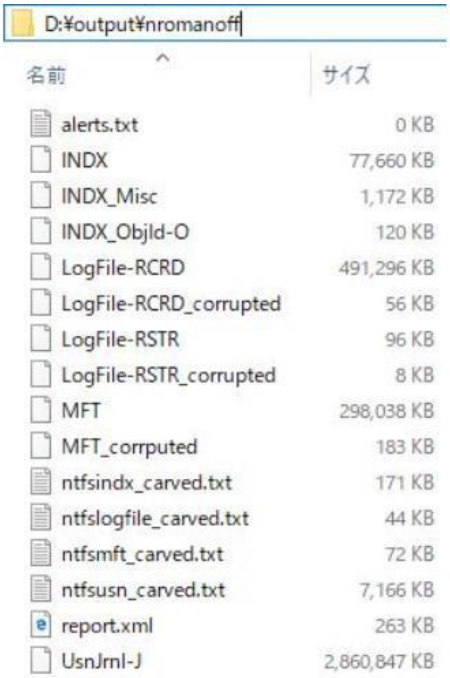
B. EXAMINING EMAIL-RELATED EVIDENCE

1. Open the acquired memory dump using the memory examination procedure specified in the lab manual.
2. Examine the available results for email-related information.
3. Open email.txt to inspect the email-related information identified by the procedure.
4. Open email_histogram.txt to examine the email-related summary or histogram output.
5. Record relevant observations from the generated files.
6. Preserve the acquired dump and analysis results for documentation.

OBSERVATION:




Run bulk_extractor
Required Parameters
Scan: Image File Raw Device Directory of Files
Image file E:¥win7-32-nromanoff-c-drive.E01
Output Feature Directory D:¥output¥nromanoff2
General Options
Use Banner File
Use Alert List File
Use Stop List File
Use Find Regex Text File
Use Find Regex Text
Use Random Sampling
Tuning Parameters
Use Context Window Size 16
Use Page Size 16777216
Use Margin Size 4194304
Use Block Size 512
Use Number of Threads 4
Use Maximum Recursion Depth 7
Use Wait Time 60
Parallelizing
Use start processing at offset
Use process range offset o1-o2
Use add offset to reported feature offsets
Debugging Options
Scanners
base16
facebook
outlook
sceadan
wordlist
xor
accts
aes
base64
elf
email
exif
find
gps
gzip
hiberfile
httplogs
json
kml
msxml
net
ntfsindx
ntfslogfile
ntfsmft
ntfsusn


D:¥output¥nromanoff
名前
サイズ
alerts.txt 0 KB
INDX 77,660 KB
INDX_Misc 1,172 KB
INDX_ObjId-O 120 KB
LogFile-RCRD 491,296 KB
LogFile-RCRD_corrupted 56 KB
LogFile-RSTR 96 KB
LogFile-RSTR_corrupted 8 KB
MFT 298,038 KB
MFT_corrputed 183 KB
ntfsindx_carved.txt 171 KB
ntfslogfile_carved.txt 44 KB
ntfsmft_carved.txt 72 KB
ntfsusn_carved.txt 7,166 KB
report.xml 263 KB
UsnJrnl-J 2,860,847 KB

RESULT:

Thus, the procedure for collecting email-related evidence from a computer using DumpIt for RAM acquisition and subsequent memory examination was studied.

EXPERIMENT 9

HIDING AND EXTRACTING A TEXT FILE BEHIND AN IMAGE/AUDIO FILE USING COMMAND PROMPT

AIM:

To demonstrate file concatenation by attaching a text file to an image and a ZIP archive to an audio file, using Command Prompt and WinRAR as specified.

SOFTWARE REQUIREMENTS:

- Operating System: Windows
- Command Prompt (CMD)
- WinRAR (for the audio file method)
- Image file (B.jpeg)
- Audio file (sound.mp3)
- Text files (A.txt and secret.txt)

THEORY:

1. Definition: Concatenation combines the contents of two or more files into a single output file.

2. First Exercise: A text file is appended to an image file using Command Prompt.

3. Second Exercise: A text file is compressed into a ZIP archive using WinRAR, and the archive is appended to an audio file using Command Prompt.

4. Data Concealment: The combined file retains the original image or audio data, with additional content placed after it.

5. Forensic Examination: Investigators examine the file structure to identify appended data that may be hidden.

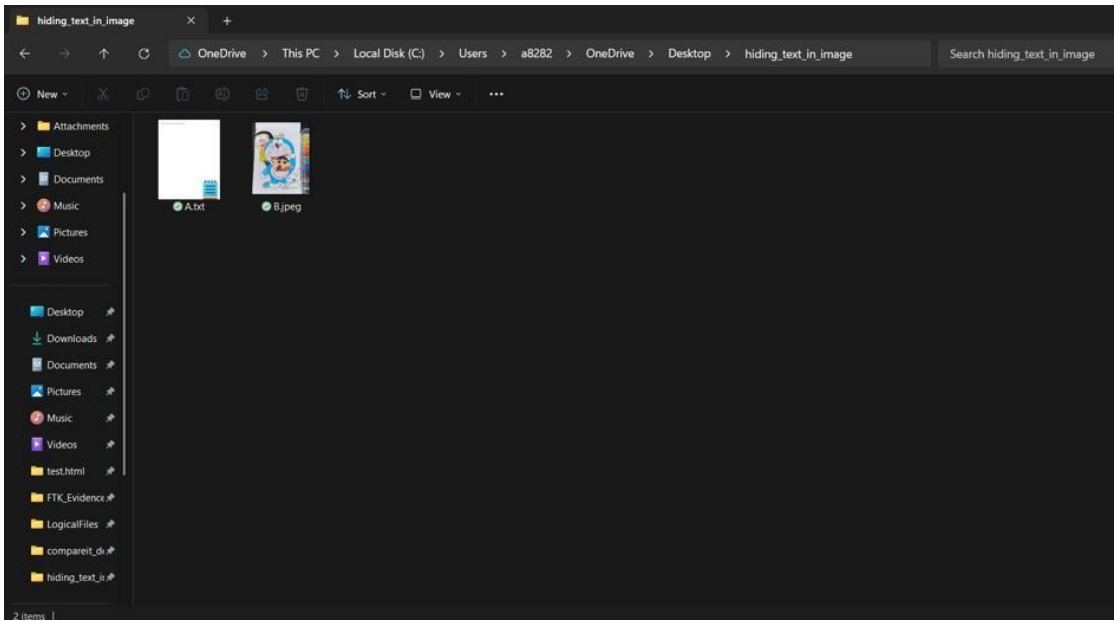
6. Data Extraction: If the appended content is identifiable and recoverable, forensic tools may help extract it.

7. Purpose: This technique demonstrates how additional data can be concealed within a file and detected through digital forensic analysis.

PROCEDURE:

A. HIDING A TEXT FILE BEHIND AN IMAGE FILE USING CMD

1. Create a folder named hiding_text_in_image on the Desktop.
2. Place the image file B.jpeg and text file A.txt inside the folder.
3. Open Command Prompt.
4. Navigate to the working directory using the following command:



Execute the following command to combine the files:

copy /b B.jpeg + A.txt C.jpeg

Where,

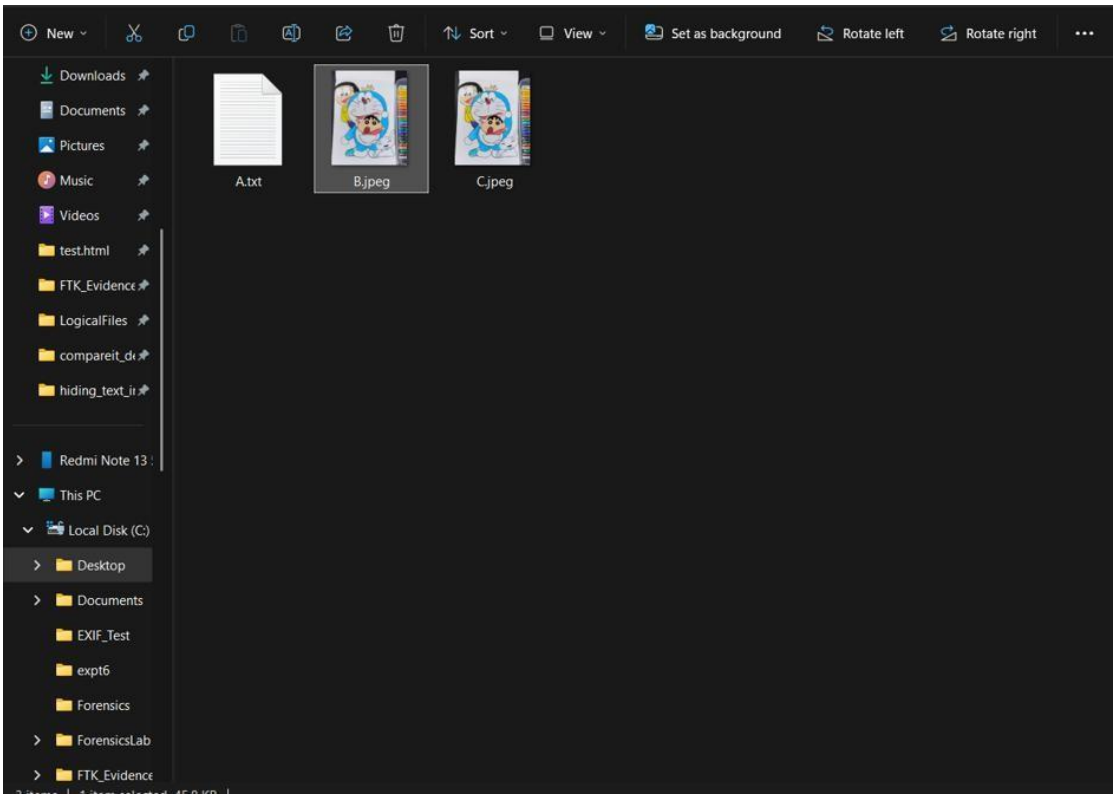
- B.jpeg — Original image file.
- A.txt — Text file to be appended.
- C.jpeg — Combined output file.
- /b — Specifies binary mode.

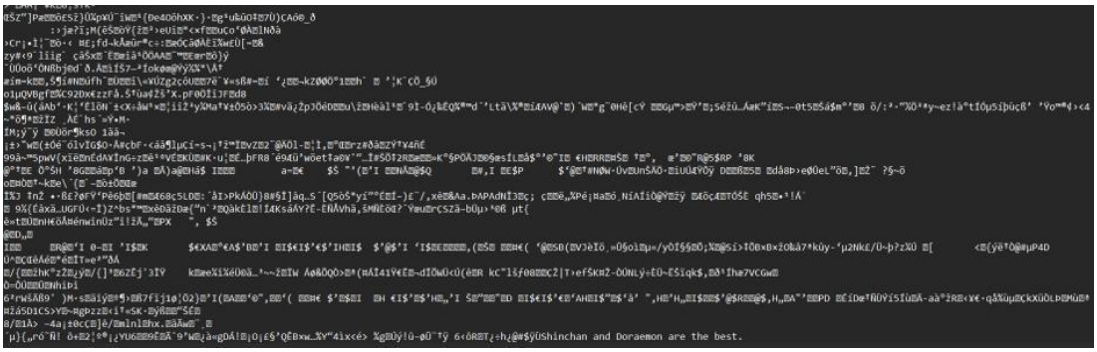
```
C:\Desktop>cd hiding_text_in_image

C:\Desktop\hiding_text_in_image>copy /b B.jpeg + A.txt C.jpeg
B.jpeg
A.txt
        1 file(s) copied.

C:\Desktop\hiding_text_in_image>
```

1. Wait until Command Prompt finishes the operation.
2. Check that C.jpeg now exists in the current folder.
3. Note the message returned by Command Prompt.





B. HIDING A TEXT FILE BEHIND AN AUDIO FILE USING WINRAR AND CMD

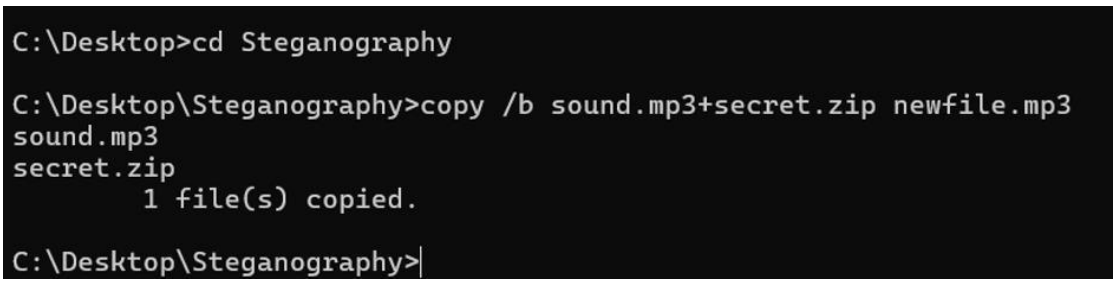
1. In Notepad, make a new text document called secret.txt.
2. Type test content and save the document.
3. Use WinRAR to compress secret.txt into an archive called secret.zip.
4. Place sound.mp3 and secret.zip together in the folder being used for the command.
5. Launch Command Prompt and change to the Steganography directory.

Execute the following command to append the ZIP archive to the audio file:

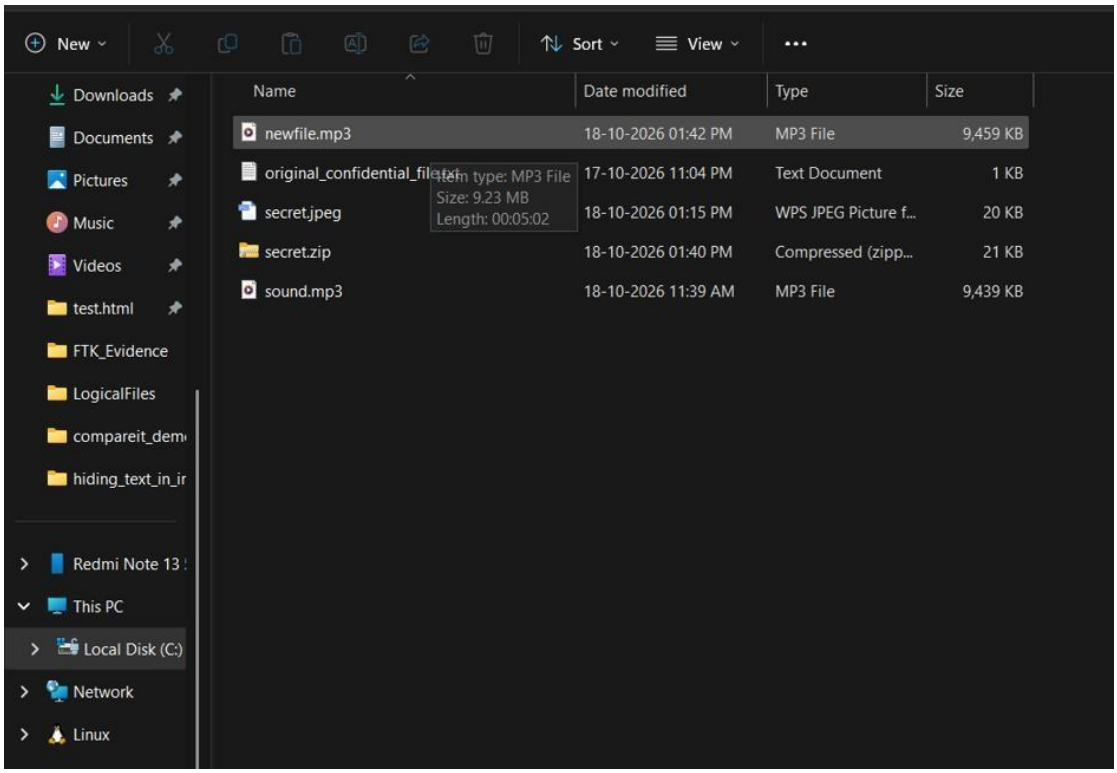
copy /b sound.mp3+secret.zip newfile.mp3

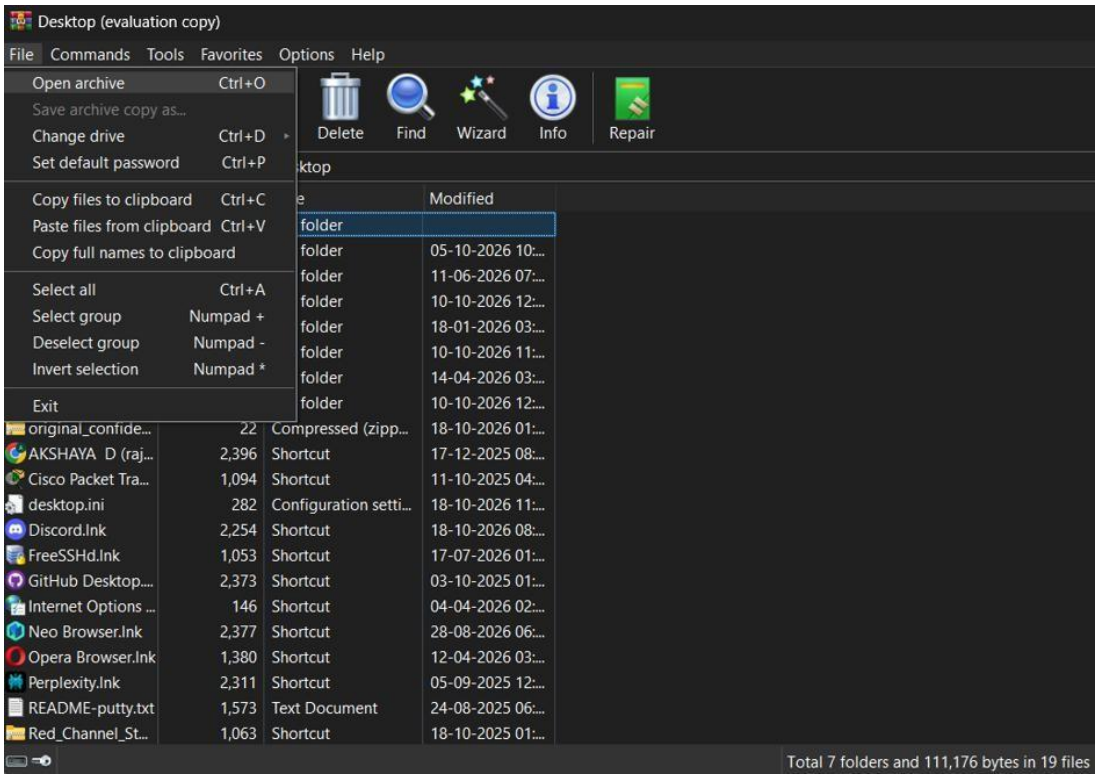
Where,

- sound.mp3 — Original audio file.
- secret.zip — ZIP archive containing the text file.
- newfile.mp3 — Combined output file.
- /b — Specifies binary mode.

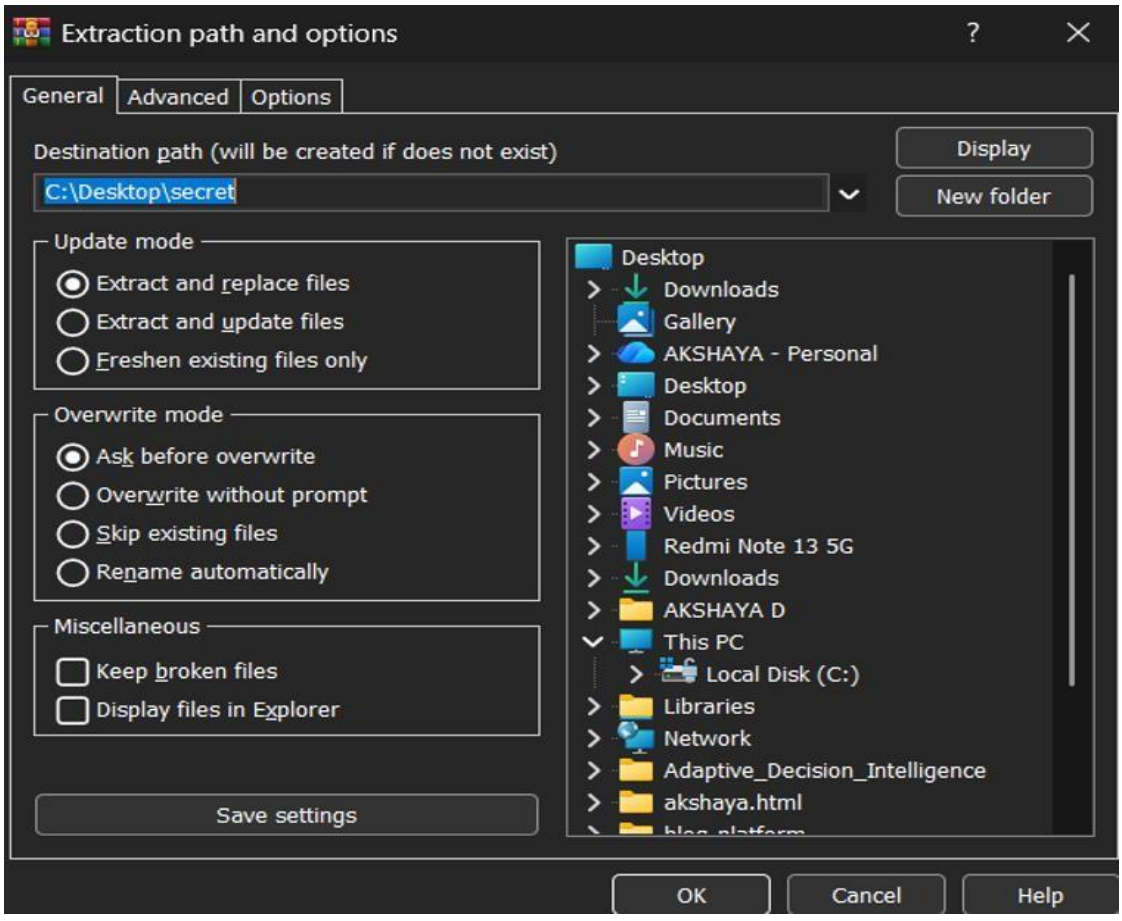


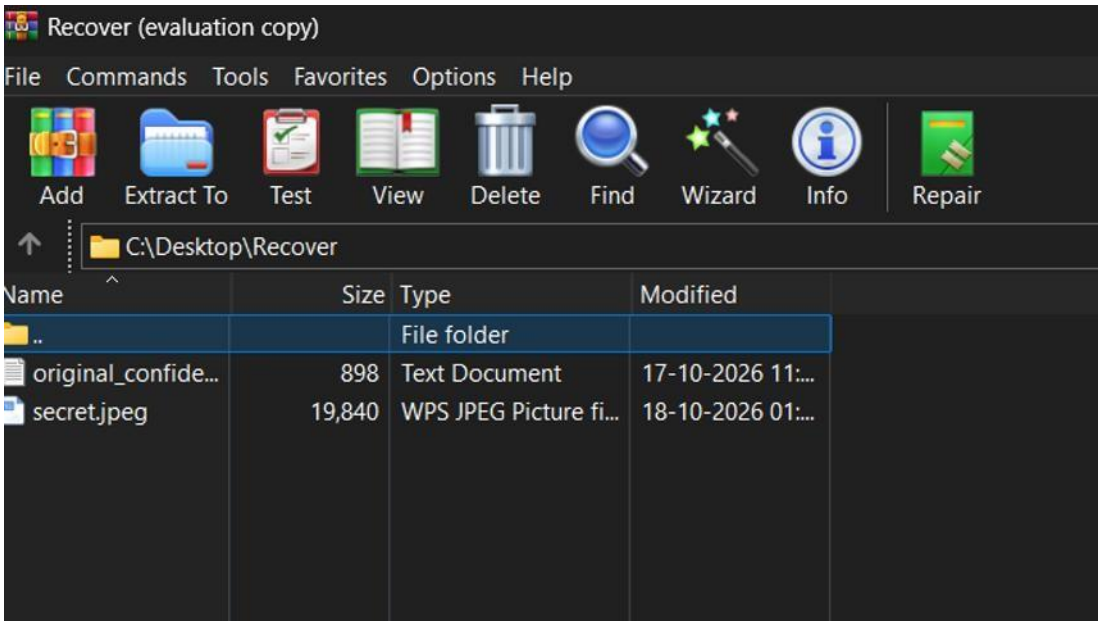
1. Wait until Command Prompt finishes the operation.
2. Confirm that newfile.mp3 appears in the working folder.
3. If needed for inspection, open newfile.mp3 in Notepad and view the displayed data.
4. To attempt recovery, make a copy of newfile.mp3 and try opening the copy using WinRAR. If the appended archive is recognized, extract its contents and verify whether secret.txt can be recovered.


New
Sort
View
Downloads
Documents
Pictures
Music
Videos
test.html
FTK_Evidence
LogicalFiles
compareit_dem
hiding_text_in_ir
Redmi Note 13
This PC
Local Disk (C:)
Network
Linux
Name
Date modified
Type
Size
newfile.mp3
18-10-2026 01:42 PM
MP3 File
9,459 KB
original_confidential_file.txt
17-10-2026 11:04 PM
Text Document
1 KB
Item type: MP3 File
Size: 9.23 MB
Length: 00:05:02
secret.jpeg
18-10-2026 01:15 PM
WPS JPEG Picture f...
20 KB
secret.zip
18-10-2026 01:40 PM
Compressed (zipp...
21 KB
sound.mp3
18-10-2026 11:39 AM
MP3 File
9,439 KB


Desktop (evaluation copy)
File
Commands
Tools
Favorites
Options
Help
Open archive Ctrl+O
Save archive copy as...
Change drive Ctrl+D
Set default password Ctrl+P
Copy files to clipboard Ctrl+C
Paste files from clipboard Ctrl+V
Copy full names to clipboard
Select all Ctrl+A
Select group Numpad +
Deselect group Numpad -
Invert selection Numpad *
Exit
Delete
Find
Wizard
Info
Repair
Modified
Total 7 folders and 111,176 bytes in 19 files


View - newfile.mp3
File Edit View Help
original_confidential_file.txt
secret.jpeg


Extraction path and options
General
Advanced
Options
Destination path (will be created if does not exist)
C:\Desktop\secret
Display
New folder
Update mode
Extract and replace files
Extract and update files
Freshen existing files only
Overwrite mode
Ask before overwrite
Overwrite without prompt
Skip existing files
Rename automatically
Miscellaneous
Keep broken files
Display files in Explorer
Save settings
Desktop
Downloads
Gallery
AKSHAYA - Personal
Desktop
Documents
Music
Pictures
Videos
Redmi Note 13 5G
Downloads
AKSHAYA D
This PC
Local Disk (C:)
Libraries
Network
Adaptive_Decision_Intelligence
akshaya.html
OK
Cancel
Help



RESULT:

Thus, a text file was appended to an image file using Command Prompt, producing C.jpeg. A ZIP archive containing a text file was also appended to an audio file using WinRAR and Command Prompt, producing newfile.mp3.

EXPERIMENT 10

EXTRACT EXIF DATA FROM IMAGE FILES USING EXIFREADER SOFTWARE

AIM:

To use ExifReader to view the Exchangeable Image File Format (EXIF) information embedded in a selected image.

SOFTWARE REQUIREMENTS:

- Operating System: Windows
- Software: ExifReader
- An image file in JPEG or JPG format

THEORY:

Exchangeable Image File Format (EXIF) is a metadata standard for image files. Cameras, phones, and other imaging devices may save details about the image and the way it was captured.

Depending on the device and file, EXIF fields may provide information such as:

- **CAMERA MAKE AND MODEL:** Information about the device used to capture the image.
- **DATE AND TIME:** The recorded date and time associated with the image, if available.
- **CAMERA SETTINGS:** Information such as shutter speed, aperture, and exposure.
- **FLASH INFORMATION:** Indicates whether the camera flash was used, if recorded.
- **IMAGE DIMENSIONS:** The width and height of the image.
- **GPS INFORMATION:** Geographic coordinates, if the device recorded them and the metadata remains available.

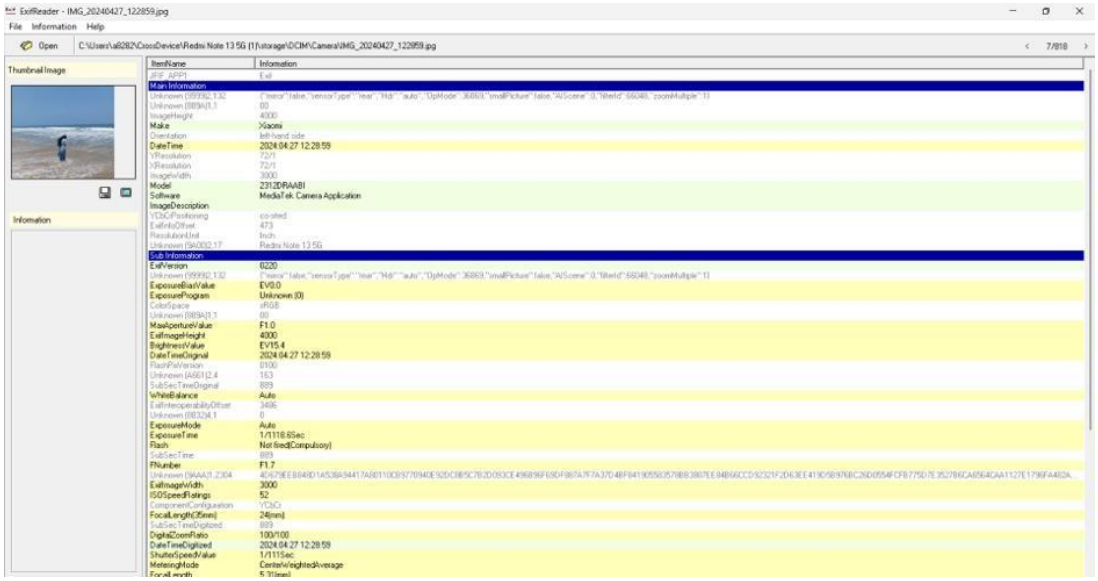
ExifReader is a Windows utility that reads and displays EXIF fields from supported image formats. The displayed metadata can be considered when examining an image as digital evidence.

PROCEDURE:

1. Obtain ExifReader and make it ready on the Windows system.
2. Run the ExifRead.exe program.
3. Select OPEN in the program window.
4. Navigate to the image file's saved location.
5. Choose a JPEG/JPG image and load it into the utility.
6. Check that the selected picture appears in the application.
7. Review the EXIF fields reported by the program.
8. Identify the available details, such as camera information, image dimensions, date and time, and camera settings.
9. Write down the metadata that is relevant to the investigation.
10. Take a screenshot showing both the image and the metadata panel.

OBSERVATION:

The selected picture was opened in ExifReader, which showed the EXIF fields that were present in the file.



RESULT:

Hence, ExifReader was used to inspect the selected image's EXIF metadata and note the information retained in the file.